

Project Report

ComicCraft - AI ComicCraft Story Creator using Gemini Models

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Project Name	ComicCraft - AI ComicCraft Story Creator using Gemini Models
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1. INTRODUCTION

1.1 Project Overview

ComicCraft is a web-based application that uses AI to generate personalized comic book stories and illustrations from user-provided prompts. It is built with FastAPI and integrates Google's Gemini models with Stable Diffusion (via Hugging Face's Diffusers library) to automate storylines, dialogues and comic-style imagery.

The user supplies a story prompt, main character name, setting, tone and art style. ComicCraft returns a panel-by-panel storyline with matching illustrations, shown in a web preview and downloadable as a PDF containing both images and narration. Both an HTML front end and API endpoints are provided, so comic creation is accessible to non-artists and casual users.

1.2 Purpose

- Let anyone turn a simple idea into a complete, illustrated comic without drawing or writing skills.
- Automate the full pipeline: panel outline, narration and dialogue, illustration, layout and PDF export.
- Give users control over mood and look (tone and art style) and let them regenerate until the comic matches their vision.
- Demonstrate how large language models and diffusion models can be combined in one end-to-end application.

2. IDEATION PHASE

2.1 Problem Statement

Creating a comic normally requires scripting, character and scene design, and illustration, which demands time and artistic skill. Hobbyists, students and storytellers with good ideas often cannot turn them into a finished visual story. There is a need for a simple tool that takes a short idea and produces a cohesive, illustrated, shareable comic automatically.

2.2 Empathy Map Canvas

Target user: a casual storyteller or hobbyist who is not an artist.

Says	Thinks
"I have a great story idea but I can't draw." "I want it to look like anime / a classic comic."	Will it match my vision and mood? Can I change the style if I don't like it?
Does	Feels
Types a short story idea, picks a character, setting, tone and style, and reviews the result.	Excited to see ideas visualised; frustrated by the effort and skill traditional comic making needs.

2.3 Brainstorming

- Split the work between models: Gemini Flash for fast structured outlines, Gemini Pro for rich narration and dialogue, Stable Diffusion for images.
- Use a fixed 5-panel structure so every comic has a clear beginning, middle and end.
- Capture five inputs (prompt, character name, setting, tone, art style) so output feels personal.
- Show a panel-by-panel web preview before export, then offer a PDF download.
- Expose both an HTML form and JSON API routes for users and developers.
- Possible later ideas: user accounts, a comic library, multi-page story arcs, cloud deployment.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User action	System response
Discover	Opens the ComicCraft home page.	Shows a clean form with a scenic background.
Create	Enters prompt, character, setting, tone and art style.	Generates outline, story and panel images.
Review	Reads the comic preview panel by panel.	Displays title, image, scene description, caption and narration.
Customize	Changes tone or style (e.g. funny, comic book) and resubmits.	Regenerates the full pipeline with new settings.
Export	Clicks "Download Your Comic as PDF".	Builds a timestamped PDF and downloads it.
Return	Lands on the export success page.	Offers "Go Create Another Comic".

3.2 Solution Requirement

Functional requirements

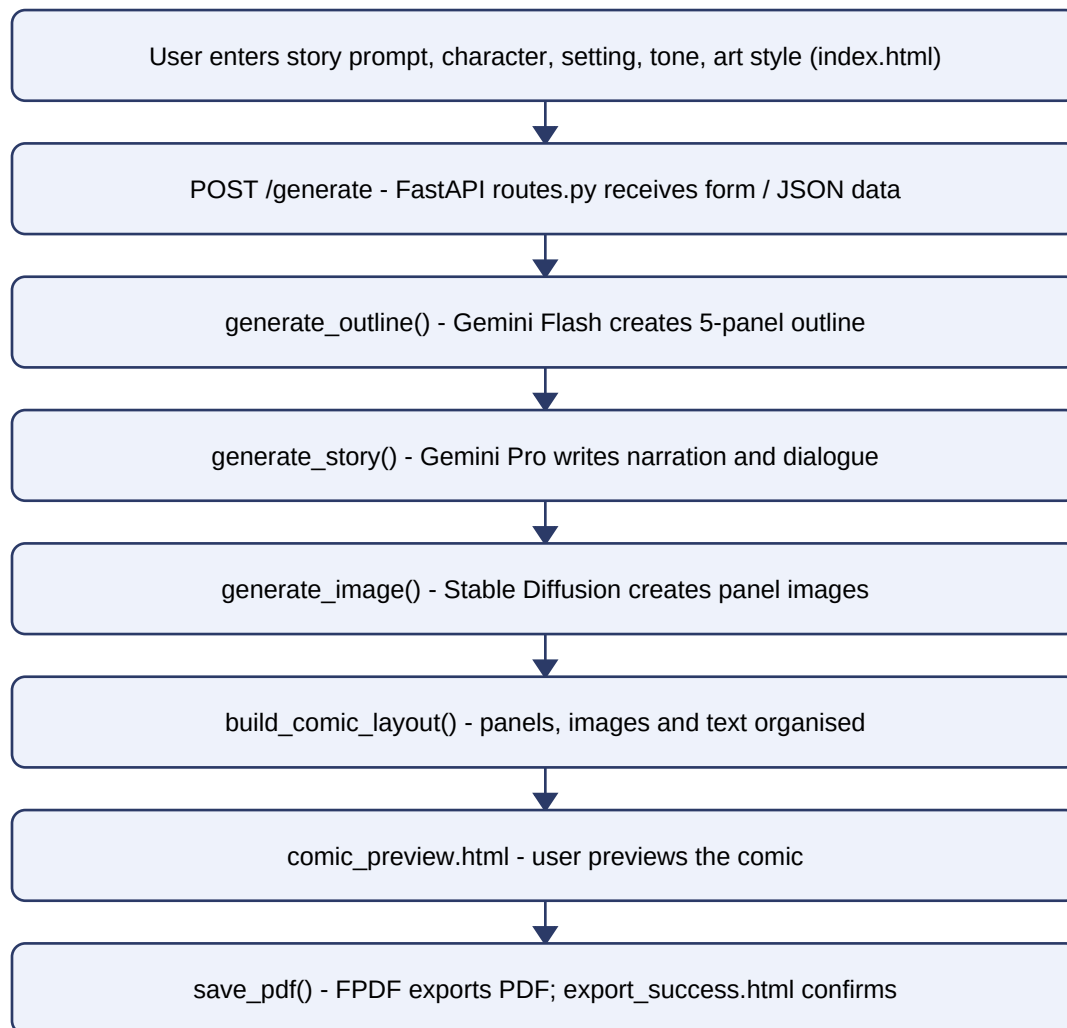
- Accept story prompt, character name, setting, tone and art style through an HTML form or a JSON request.
- Generate a structured 5-panel outline (panel number, title, scene description, image prompt).
- Generate narration and character dialogue for each panel.
- Generate a comic-style image for each panel and save it under static/panels.

- Assemble panels into a layout and show a preview page.
- Export the comic as a multi-page PDF saved in static/exports and confirm on a success page.

Non-functional requirements

- Simple, responsive and easy-to-use interface.
- Modular code (separate files for each AI function, layout and export).
- Secure handling of API keys through environment variables.
- Graceful error handling for API and JSON failures.

3.3 Data Flow Diagram



3.4 Technology Stack

Layer	Technology
Backend	Python, FastAPI, Uvicorn (ASGI server), Pydantic
Frontend	HTML, CSS, Jinja2 templates
Text AI	Google Gemini 1.5 Flash (outline) and Gemini 1.5 Pro (story) via google-generativeai
Image AI	Stable Diffusion (runwayml/stable-diffusion-v1-5) using Hugging Face Diffusers, Transformers, PyTorch
Export	FPDF, Pillow
Tools	pip, virtualenv, python-dotenv, Git

4. PROJECT DESIGN

4.1 Problem Solution Fit

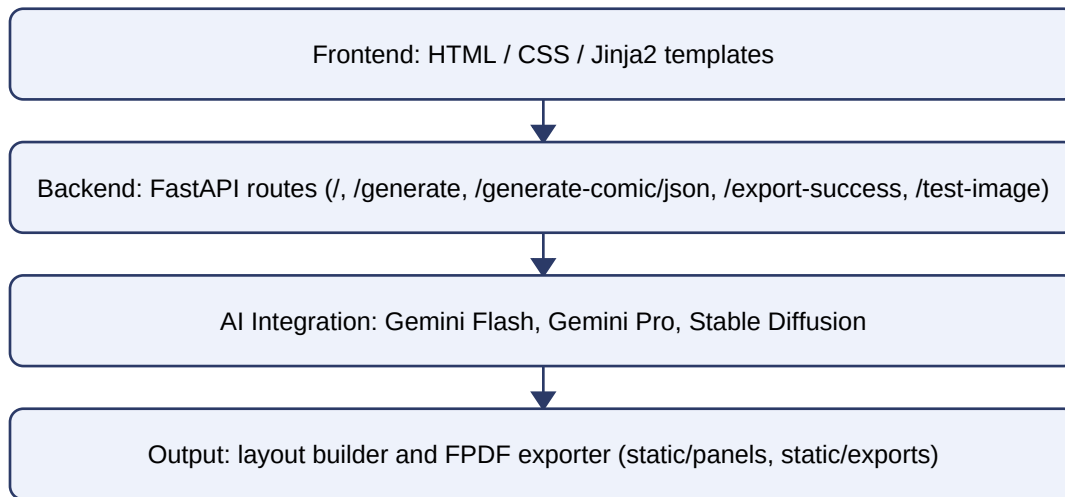
Problem	ComicCraft solution
Comic making needs drawing and writing skill.	AI writes the story and draws the panels from a short prompt.
Results rarely match the creator's mood or style.	Tone and art style selections drive prompts; users can regenerate.
Hard to share or keep the result.	One-click PDF export with images and narration.
Multiple tools needed for story, art and layout.	Single web app handling the whole workflow.

4.2 Proposed Solution

A FastAPI web application that chains three AI steps. Gemini Flash converts the user's prompt into a structured 5-panel outline. Gemini Pro expands the outline into narration and dialogue. Stable Diffusion generates an illustration for every panel using the image prompts from the outline. The backend arranges everything into a layout, renders a preview with Jinja2 templates, and exports a PDF with FPDF.

4.3 Solution Architecture

The system has three components: the **Frontend** (index.html, comic_preview.html, export_success.html), the **Backend** (FastAPI with routes.py, layout_builder.py, exporters.py), and **AI Integration** (gemini_flash.py, gemini_pro.py, image_generator.py).



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Milestone	Activities	Outcome
1. Model selection and architecture	Research and select models; define architecture; set up environment.	Gemini Flash, Gemini Pro and Stable Diffusion selected; 3-part architecture defined.
2. Core functionalities	Build generate_outline, generate_story, generate_image, build_comic_layout, save_pdf; implement FastAPI backend.	Complete generation pipeline.
3. routes.py development	Write route handlers for form and JSON input, AI calls and error handling.	Routes / , /generate, /generate-comic/json, /export-success, /test-image.
4. Frontend development	Design UI; create Jinja2 templates.	index, comic_preview and export_success pages.
5. Deployment	Prepare and test local deployment with Uvicorn.	App running at http://127.0.0.1:8000 with API docs at /docs.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The application was run locally with `uvicorn app.main:app --reload` and each workflow was exercised through the web interface and the FastAPI docs page.

Test case	Expected result
Open home page (/)	Input form loads with all five fields.
Submit form to /generate	Outline, story and 5 panel images are generated and the preview page opens.

Test case	Expected result
Change tone to “funny” and art style to “comic book”	Pipeline reruns and the comic reflects the new mood and look.
POST JSON to /generate-comic/json	Returns layout data and the PDF path.
Call /test-image with a custom prompt	A single image is generated and saved.
Click Download as PDF	Timestamped PDF is saved and the export success page appears.
Invalid input or API failure	Handled with HTTPException / try-except without crashing.

7. RESULTS

7.1 Output Screenshots

[Insert screenshot: Home page (index.html) - story input form]

[Insert screenshot: Comic preview page (comic_preview.html) - generated panels]

[Insert screenshot: Download as PDF button and exported PDF]

[Insert screenshot: Comic export success page (export_success.html)]

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Creates complete illustrated comics from a short prompt.	Currently single-comic generation with no user accounts or saved library.
Needs no drawing or writing skill.	Fixed 5-panel length.
Customisable tone, setting and art style.	Needs Gemini and Hugging Face credentials and internet access.
Modular design and both UI and API access.	Image generation can be slow on modest hardware.
Professional PDF export for saving, printing and sharing.	Deployed locally only for now.

9. CONCLUSION

ComicCraft shows the creative potential of AI-driven comic generation. By combining Gemini Flash and Gemini Pro for storytelling with Stable Diffusion for illustration, it turns simple prompts into complete, visually engaging comics, from outline and narration to images and a downloadable PDF, in an intuitive web interface. It simplifies comic creation for non-artists, storytellers and hobbyists and shows how AI can democratize storytelling.

10. FUTURE SCOPE

- User profiles and authentication.
- A comic library to save, revisit and share comics.
- Multi-page story arcs and variable panel counts.
- Deployment to a cloud platform.
- More art styles, characters and language options.

11. APPENDIX

Source Code

Project structure: routes.py (routing), gemini_flash.py, gemini_pro.py, image_generator.py, layout_builder.py, exporters.py, main.py; templates index.html, comic_preview.html, export_success.html; output folders static/panels and static/exports.

Dataset Link

Not applicable. The project uses pre-trained models (Gemini 1.5 Flash, Gemini 1.5 Pro and runwayml/stable-diffusion-v1-5) and no custom dataset.

GitHub & Project Demo Link

GitHub: [add repository link]

Demo: [add demo video link]